9. REFERENCE INDEX (FULL-SOURCE)

9.1 FUNDAMENTAL PHYSICAL AND KINETIC MODELS

[1] High-Energy Fluidic Processing: Domain foundation encompassing thermodynamic and kinetic optimization of multi-phase liquid streams. Math: Reynolds Number (Re=rho*v*D/mu). Utility: Essential for maintaining stable laminar flow (Re<2000).

[2] Plasma-Facing Materials Science: Isotope Retention and Damage Mechanics. (Causey, 2002). Math: S = K*sqrt(P) (Sieverts' Law). Utility: Validates the material degradation limits and hydrogen retention kinetics of the refractory matrix (Layer B).

[3] Advanced Thermal-Hydraulic Engineering: Liquid Metal Convective Heat Transfer. (Pioro et al., 2004). Math: Nu=h*D/k (Nusselt Number). Utility: Provides empirical validation for convective heat transfer rate in the dynamic stripping shell (Layer C).

[4] Continuous-Flow Metamaterial Synthesis: Continuous-Flow Mass Transport. (Free, 2013). Math: dc/dt + v*nabla c=D*nabla^2 c. Utility: Supports the feasibility of the continuous-flow waveguide and the recirculating fluidic quench bath over stagnant-batch modes.

[5] System for Multivariate Phase Convergence: High-Energy Kinematics and Conservation Laws. (Lamarsh). Math: Sum(Zi)=Ztarget. Utility: Anchors the core convergence control logic for high-density metallurgical refinement.

[6] Acoustic Cavitation Focal Point: Bubble Dynamics. (Brennen, 1995). Math: R(d^2R/dt^2)+3/2(dR/dt)^2= 1/rho(P(t)-P_inf). Utility: Validates the use of phased acoustic arrays for perfectly symmetric focal point nucleation.

[7] Refractory Ion-Sequestration & Hydrogen Mitigation: Interstitial Trapping. (Morita, 1989; Dillon, 1997). Math: D=D0*exp(-Ea/RT). Utility: Supports the active ion-sequestration function of the Layer B graphite lattice (50nm-500nm pores).

[8] Liquid-Metal Thermal Quenching & Leidenfrost Mitigation: High-Velocity Convective Stripping. (Pioro et al., 2004). Math: q=h*Delta T. Utility: Validates the dynamic convective stripping architecture (Layer C) to keep localized thermal input below the CHF threshold.

9.2 FAILURE MODE ANALYSIS

[9] Stagnant-Volume Constraints (Failure Mode): Physical limitation of stationary-bath cooling where heat buildup outpaces the fluid's specific heat dissipation capacity.

[10] Passive Buffers (Failure Mode): Structural inadequacy of solid-state Plasma-Facing Components (PFCs) under instantaneous energy spikes.

[11] Energy Dissipation Failures (Failure Mode): Mechanical and thermal limitations of static reactor architectures.

[12] Exothermic Transients (Failure Mode): Documentation of uncontrolled, high-energy particle outputs resulting from acoustically-induced lattice compression. (Taleyarkhan et al., 2002). Math: E=1/2 mv^2.

[13] Simultaneous Failure Cascades (Failure Mode): Coupled occurrence of HIC and Leidenfrost quenching (Buxbaum, 1993).

[14] Hydrogen Solubility (Failure Mode): Saturation limit of hydrogen absorption routinely exceeded in high-flux acoustic cavitation. (Causey, 2002). Math: S=K*sqrt(P) (Sieverts' Law).

[15] Hydrogen-Induced Cracking (HIC) (Failure Mode): Brittle fracture mechanics resulting from high-pressure hydrogen recombination within grain boundaries. (Callister, 2007).

[16] Proton Implantation (Failure Mode): Ballistic damage mechanism distinct from simple diffusion. (Causey, 2002). Math: -dE/dx (Stopping Power).

[17] Leidenfrost Quench Failure (Failure Mode): Rapid transition from nucleate to film boiling (Carey, 1992).

[18] CHF (Critical Heat Flux) Threshold (Failure Mode): Breakdown point of cooling efficiency in stagnant interfaces (Kraus, 2001).

[19] Anomalies in Cavitation: Thermal spiking/sonoluminescence (Flannigan & Suslick, 2005). Math: T_max = P_max*T_0/P_0.

[20] Inadequacy of Linear Models (Failure Mode): Brennen, 1995. Math: Non-linear term Phi(h,k).

[21] Thermal Runaway/DNB Crisis: Departure from Nucleate Boiling (Tong & Tang, 1997).

[22] Trapped 'Thermal Coals' (Failure Mode): Localized heat-storage reservoirs (Suslick, 1989). Math: Q= mc*Delta T.

[23] Experimental Asymmetric Bubble Collapse (Failure Mode): Lauterborn & Bolle, 1975. Math: Bjerknes forces.
[24] Microjet-Induced Fatigue (Failure Mode): Philipp & Lauterborn, 1998. Math: v_jet=sqrt(2*Delta P/rho).
[25] Synchronous Structural Collapse (Failure Mode): Collective structural loading (Taleyarkhan et al., 2002).

9.3 CORE TECHNICAL INNOVATIONS

[26] Non-linear Pressure Topology: Requirement for continuous-flow and spatial-computing frameworks. Math: Adaptive Phase-Correction Algorithm (Phi). Utility: The core technical argument supporting the necessity of the Distributed Sensory-Computational Manifold.
[27] Kelvin Force Physics: Landau & Lifshitz, Electrodynamics of Continuous Media. Utility: Validates diamagnetic separation.
[28] Microfluidic Sorting: G. M. Whitesides, Microfluidic Sorting in High-Gradient Fields. Utility: Validates mid-flight streamline migration.
[29] LIS Embodiment: W. Lauterborn, 2005. Sub-nanosecond laser-induced cavitation. Utility: Validates the alternative laser-induced shock actuation.